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Research Paper

Data-driven hydraulic property analysis and prediction of two-dimensional random fracture networks

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ABSTRACT

The large parameter space and uncertainty of the discrete fracture networks (DFNs) make the hydraulic characteristics have multiple possibilities. Statistical analysis of hydraulic characteristics based on a large amount of data is a standard practice and central to DFN methodology. The present study developed a two-dimensional (2D) DFN numerical seepage model based on Matlab and Comsol from modeling and parameter calculation to simulation to investigate the influence of parameters involved fracture network on seepage characteristics of DFNs. In total, 1728 2D DFN models were generated with increasing fracture density, fracture length, power-law exponent, fracture orientations, and fracture included angles, and $1728 \times 3 = 5184$ numerical simulations of fluid flow along the *x*-axis, *y*-axis, and center through the DFN models are conducted using a self-developed numerical code. The influence of fracture geometric features on connectivity and permeability are estimated, respectively. And to realize the prediction of permeability tensor by known geological data, a theoretical model based on the Snow model and a data mining model based on the back-propagation model are discussed. The analysis helps researchers and engineers can make informed decisions and develop effective strategies for addressing the challenges posed by uncertainty hydraulic property of the fractured rock masses.

1. Introduction

Fractures such as joints, faults, and bedding planes are ubiquitous in crustal rocks. These natural discontinuities form a complex fracture network and dominate the geomechanical and hydrological behavior of subsurface rocks with low matrix permeability. Understanding of the hydraulic property in fractured rock masses is important for the safety and performance assessment of many engineering geological applications, such as hydrogeologic surveying, geological disposal of nuclear waste, exploration of petroleum reservoirs, geologic storage of carbon dioxide, preventing groundwater contamination, and fracture grouting (Bonneau and Stoyan, 2022; Lei et al., 2014). Therefore, the analysis of hydraulic property within a fracture network has been a critical and active research topic in the subject of underground engineering. By comprehending the intricate behavior of seepage characteristics through fractures, researchers and engineers can make informed decisions and develop effective strategies for addressing the challenges posed by fractured rock masses.

With the development of a fracture characterization model (Alghalandis, 2017; Zhu, et al., 2018), continuous efforts have been made for correlating the fracture geometries and rock permeability by a series of analytical models, numerical models, and laboratory tests. For instance, the influence of fracture length with a power law or lognormal distribution on the equivalent permeability for fractured rocks were studied (De Dreuzy et al., 2001, 2012; Klimczak et al., 2010). The order of the power-law has been proven to profoundly impact fluid flow and contaminant transport behavior within a network (Revvels et al., 2013, 2023). The influence of the aperture correlated and uncorrelated fracture length on permeability were demonstrated (Chen, 2020; Baghbanan and Jing, 2007) and confirmed the importance of the correlated square root relationship of the aperture to length scaling. The impact of horizontal spatial clustering of fracture networks was illustrated to exert a significant impact on water flow and solute breakthrough times and retention (Akara et al., 2021). Moreover, to effectively evaluate the permeability, conductivity parameters related to the statistical geometrical parameter of fracture networks were defined. Oda (1986)

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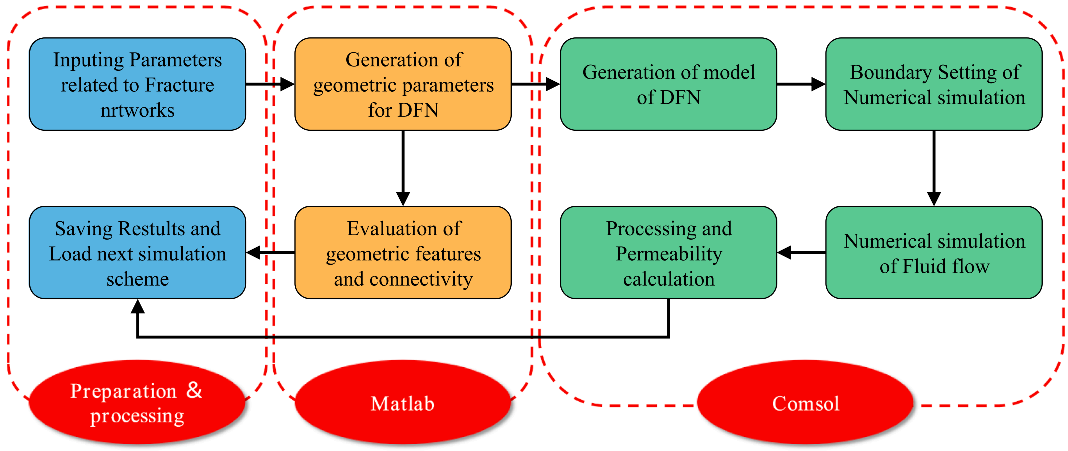


Fig. 1. Two-dimensional fracture network seepage program framework.

proposed index F_0, considering the fracture length and density based on the equivalent continuum model. Leung and Zimmerman (2012) introduced a correction coefficient η to characterize the equivalent permeability coefficient, compensating for the shortcomings in predicting the permeability of discontinuous fracture networks. Ye et al. (2021) and Bianchi and Pedretti (2017) used the geological entropy as an index to solve the spatial disorder of fracture networks. Zhu et al. (2021) demonstrated the impact of fracture geometry and topology on the connectivity and flow properties of networks. Spontaneously, faced with more complex fracture conditions, many self-developed software have been developed to simulate the seepage process of fluids in fracture networks based on discrete fracture models and equivalent fracture models. DFNWorks and FracMan are the most commonly used software developed by scholars (Hyman and Jimenez-Martinez, 2018; Dershowitz, 2014). Furthermore, Chen et al. (2015) proposed a multiple boundary upscaling method and investigated how fracture geometric properties affect the permeability of highly heterogeneous. Zou et al. (2023) developed a channel network model simplification the three dimensional fracture network and parameterized it using geological and hydraulic test data. Huang et al. (2022) proposed a three-dimensional fracture network model and studied the fracture parameters and permeability characteristics within rough fracture networks. Considering the coupled geomechanics and fluid flow, Lei (2016) developed the finite discrete element method and found the effects of geomechanical changes on the validity of a DFN and fluid flow channels. Meanwhile, to better predicate the permeability of DFNs, the empirical equation, analytical and numerical approaches were used (Shahbazi et al., 2020; Mourzenko et al., 2011). Empirical models concerning the correlation between geological indices and permeability were developed using data obtained from in situ field tests (Louis, 1972; Chen et al., 2018). Analytical methods related to permeability were developed by taking into account the geometrical characteristics of the joint sets and coupling effect (Zhou et al., 2008; Lang et al., 2014; Yao et al., 2020). And more complex hydraulic connectivity models considering anisotropy, flow path were investigated by numerical methods (Xiong et al., 2019; Zhu, et al., 2021; Zhou, et al., 2024).

However, owing to the large parameter space and uncertainty of the fracture network, even under the same statistical parameters, there are multiple possibilities for the permeability characteristics of the fracture network. Thus, the use of multiple realizations is a standard practice and central to DFN methodology. Running a large number of numerical simulations and multiple realizations for each case poses a challenge. Additionally, fractured networks exhibit high heterogeneity and anisotropy. The symmetric full tensor form is commonly used to characterize rock permeability. However, in field tests, fluid diffuses radially from the central borehole to the surrounding area, such as fracture grouting in mining or tunnel engineering, pumping tests, dewatering tests, or enhanced geothermal systems. The parameters obtained through on-site measurement, which include injection pressure, permeability, specific water absorption, and grouting amount, are different from the permeability k_{xx} and k_{yy} in direction x or y, which include injection pressure, permeability, specific water absorption, and grouting amount. Thus, a second challenge is to explore whether it is possible to establish an effective prediction relationship or model for permeability tensor based on typical hydrogeological data and fracture parameters to understand the diffusion range and shape of fluid flow in networks.

Specifically, our study aimed to the following: (1) Propose a method for conducting large-scale numerical simulations, simulating the permeability characteristics of 1,728 fracture networks and 1,728 * 3 = 5184 flow scenarios; (2) investigate the influence of fracture geometry parameters on the density, connectivity, and permeability of the fracture networks under a large number of numerical simulations; and (3) combine the known geological data and inflow rate obtained from single-hole injection tests to predict the permeability tensor in discontinuous fracture networks.

The remainder of the paper is organized as follows. Section 2 gives a description of the developed program and DFN model and introduces the method for fracture network conductivity and numerical simulation. Section 3 analyzes the numerical simulated data of the relationship between permeability, fracture network, and connectivity. Section 4 proposes the prediction model of the permeability tensor. Finally, a discussion is given in Section 5, and conclusions are drawn in Section 6.

2. Methodology

To run a large number of numerical simulations and multiple realizations for each case, a complete two-dimensional (2D) fracture network seepage program was developed based on the secondary development of Matlab and Comsol Multiphysics in this study. The fully automatic cycle calculation is realized by inputting all DFN designs and followed by the fracture geometric parameters generation of 2D DFN → evaluation of geometric features and connectivity → 2D DFN modelling → numerical simulation boundary setting → seepage simulation → processing and saving of simulation results. This significantly reduces the manual operation time, improves the computational efficiency of numerical simulation, and makes it possible to conduct extensive slurry diffusion numerical simulations. The specific workflow is shown in Fig. 1. More in-depth discussion of the above mentioned methodologies and a detailed discussion of these five parts are provided in respective subsections.

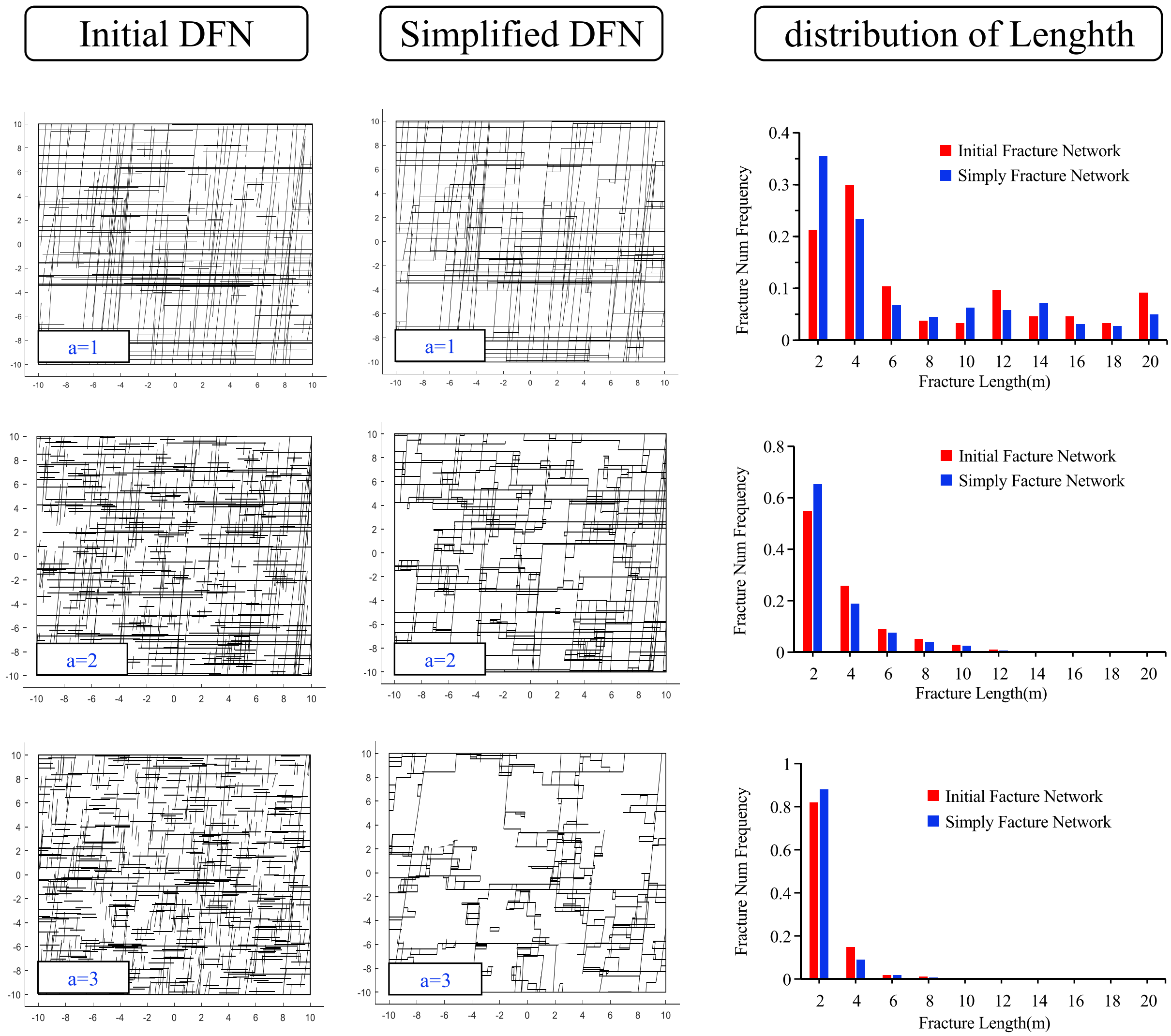


Fig. 2. Initial and simplified fracture networks with different power-law exponents *a*.

Table 1
Fracture network parameters for different length exponents *a* and fracture count densities.

a	Fracture Count density	Initial Fracture Network				Simplified Fracture Network			
		Fracture Number	P_{12}	I_C	η	Fracture Number	P_{12}	I_C	η
1	0.3	240	4.16	7.46	166	232	3.61	7.91	148.03
2	0.7	560	3.91	2.79	100.26	468	2.54	3.34	70.48
3	1	800	3.49	1.39	67.77	434	1.41	2.04	31.9

2.1. Discrete fracture model generation

Fractured rock permeability is determined by the geometry of the fracture network when the rock has an impervious or weakly permeable matrix (Zou and Cvetkovic, 2021). However, full characterization of fractured rock masses is not possible owing to a limited number of known fracture characteristics and extremely small sample of the overall fracture network. To generate representative, site-specific fracture networks, field campaigns play a crucial role in characterizing fractured rock masses, utilizing various sources such as boreholes (e.g., electrical resistivity, ultrasonic, and optical televiewers), rock outcrops, seismic images, and hydraulic tests. Furthermore, the stochastic DFN modeling approach proposed by field investigation and statistical analysis has made significant contributions in establishing fracture networks (Alghalandis, 2017; Forstner and Laubach, 2022).

Statistical properties of fracture parameters, including fracture location, length, orientation, aperture, and spacing, can be described using specific probability distributions. In this study, fracture centers are assumed to conform to the Poisson distribution to reflect clustered fracture spacing (Akara et al., 2021). Fracture orientations are modeled by the Fisher distributions to constrain orientation deviation on a unit sphere (Mardia and Jupp, 2000). For fracture length, Bonnet et al. (2001) conclusively determined that fracture length is a power-law distributed variable by the largest dataset of fracture length to date, which can be expressed as $n(l) = Al^{-a}dl$, where power-law exponent a represents fracture growth properties, typically ranging from 1 to 3. As shown in Fig. 2(a–c), fracture networks generated with $a = 1$ produce backbone structures dominated by long fractures. The proportion of fracture lengths exceeding 2 m is 0.78. Conversely, fracture networks with $a = 3$ produce backbone structures dominated by short fractures.

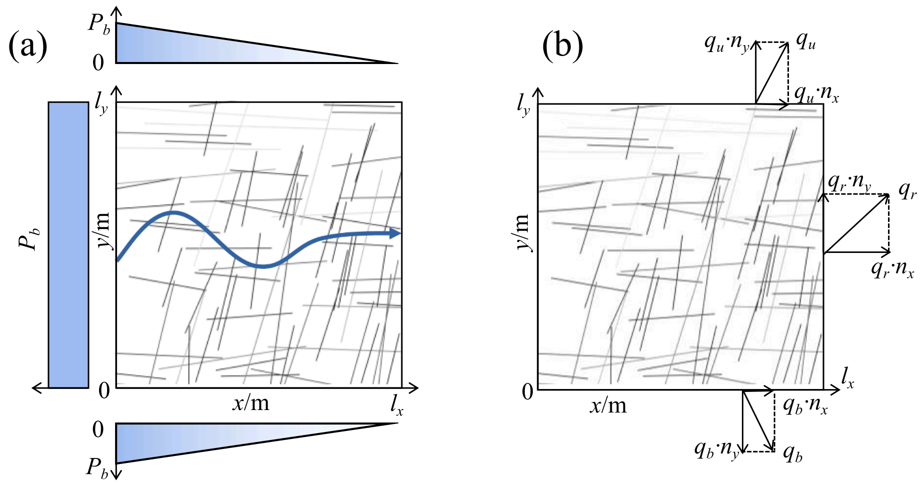


Fig. 3. Finite element model of fracture seepage: (a) seepage simulation under linear boundary conditions along the x-axis, (b) Flow rate calculation diagram.

Table 2
Simulation scenarios of 1728 DFN models.

Factors	Level
Fracture set	2
power law exponent a	1; 2; 3

Fracture count density ρ	a = 1, ρ = [0.4, 0.6, 0.8, 1.0];
	a = 2, ρ = [0.6, 0.8, 1.0, 1.2];
	a = 3, ρ = [0.8, 1.0, 1.2, 1.4] ;

Minimum Fracture Length $L_{\min}$	1.5 m; 2 m; 2.5 m; 3 m
Maximum Fracture Length $L_{\max}$	20 m
Fracture orientations α	0°; 22.5°; 45°
Fracture included angles θ	15°; 40°; 65°; 90°

The proportion of lengths less than 2 m is 0.82. In contrast, fracture networks with $a = 2$ exhibit backbone structures dominated by a combination of long and short fractures. The proportion of lengths exceeding 4 m is 0.45. Statistical analysis (Table 1) also reveals that despite an increase in fracture count density, both fracture length density P_{12} (m/m^2) and the connectivity index exhibit a decrease as power-law exponent a increases in the initial fracture network. This observation aligns with previous studies, which have demonstrated that the exponent of the power-law a plays a significant role in determining network connectivity and permeability (De Dreuzy et al., 2012; Reeves et al., 2013; 2023). Thus, the fracture length with a power-law distribution was adopted to characterize the length in this study. Fracture aperture can be constant or correlated with fracture length. A power-law relationship between fracture aperture and length is widely recognized at the field scale (Renshaw and Park, 1997) and derived from linear elastic fracture mechanics (Olson, 2003). It can be expressed as $w = \gamma l^D$ (Schultz et al., 2008), where w denotes fracture aperture, l represents fracture length, γ is the coefficient associated with fractured rock, and correlation exponent D indicates the mechanical interaction between closely spaced fractures, ranging from 0.5 to 1.0. In this study, the aperture correlated with fracture length was used, D was set to 0.5, and γ was assumed to be 1.2×10^{-4} based on field data (Schultz and Soliva, 2012). Furthermore, Fig. 2 shows the initial fracture networks and the equivalent fracture after removing isolated and dead-end fractures. The simplification of fracture network can also profoundly affect the parameters of fracture density P_{12} and connectivity I_C and η. To demonstrate the relationship between real network parameters and permeability, all fracture networks are simplified by removing end points and isolated fractures.

2.2. Fracture density and connectivity characteristic

2.2.1. Fracture density

In the 2D fracture network model, the fracture density is represented by fracture length density P_{12}, which is defined as the total length of fractures per unit area and can be expressed as:

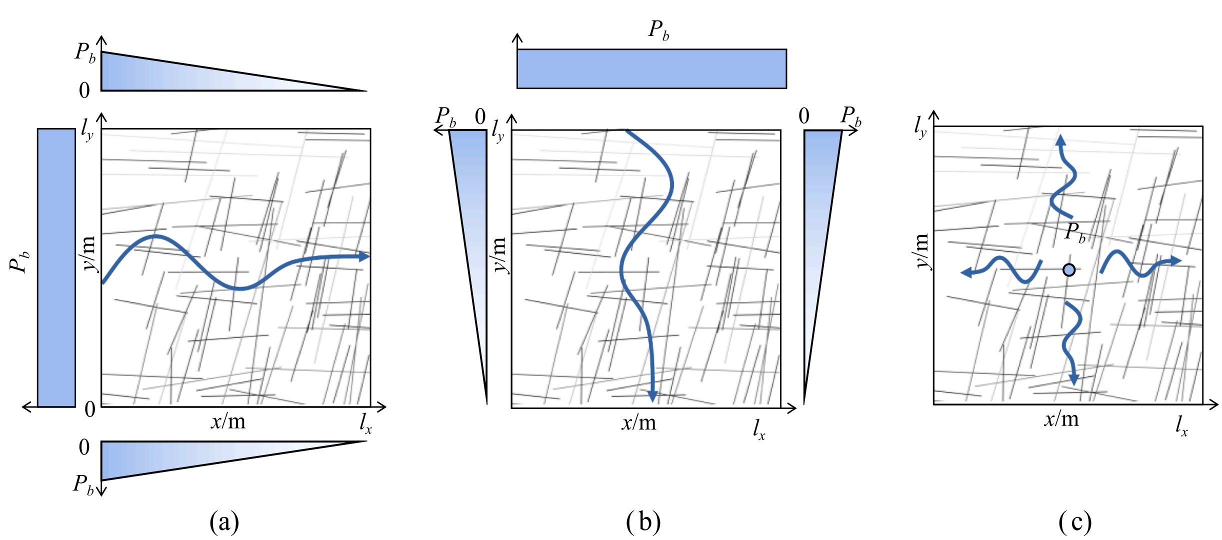


Fig. 4. Boundary conditions of numerical simulations of fluid flow along different direction: (a) x-axis, (b) y-axis, (c) center radial diffusion.

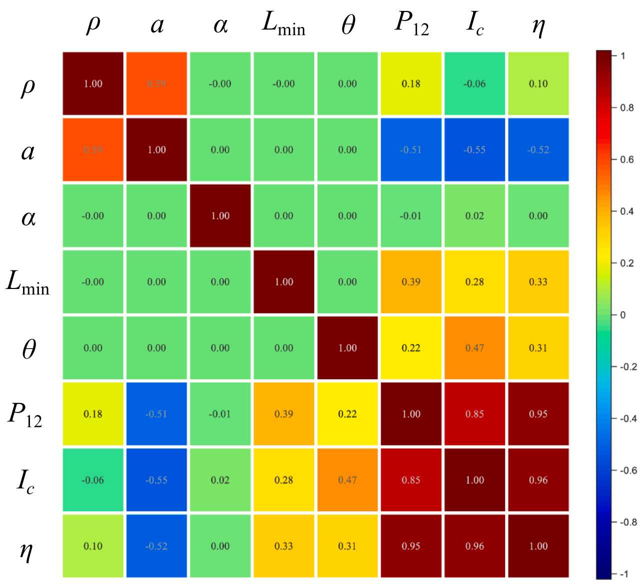


Fig. 5. The heat map of PPC between the fracture geometric features and the connectivity index.

$$P_{12} = \sum_{i=1}^{j} L_i \Big/ A \tag{1}$$

where L_i is the fracture length of i-th fracture, A is the domain area.

2.2.2. Connectivity

Given the extreme complexity of fracture networks, connectivity evaluation serves as an effective means to describe the flow paths and mechanical behavior of underground rock masses. Scholars have employed various descriptive methods to characterize networks connectivity.

Connectivity index I_c is a common indicator to describe the degree of fracture connectivity. I_c has the strongest correlation with fracture network permeability in fracture-dominated geothermal reservoirs (Jafari and Babadagli, 2011, 2013). It is defined as the ratio of the number of intersection points to the total number of fracture lines, which can be expressed as:

$$I_c = \frac{N_{node}}{N_F} \tag{2}$$

where N_{node} is the number of intersection points in fracture networks, and N_F is the number of fractures in dominate domain area.

Connectivity index η, as a conductivity parameter, represents the combined measure of both the length and connectivity of a fracture

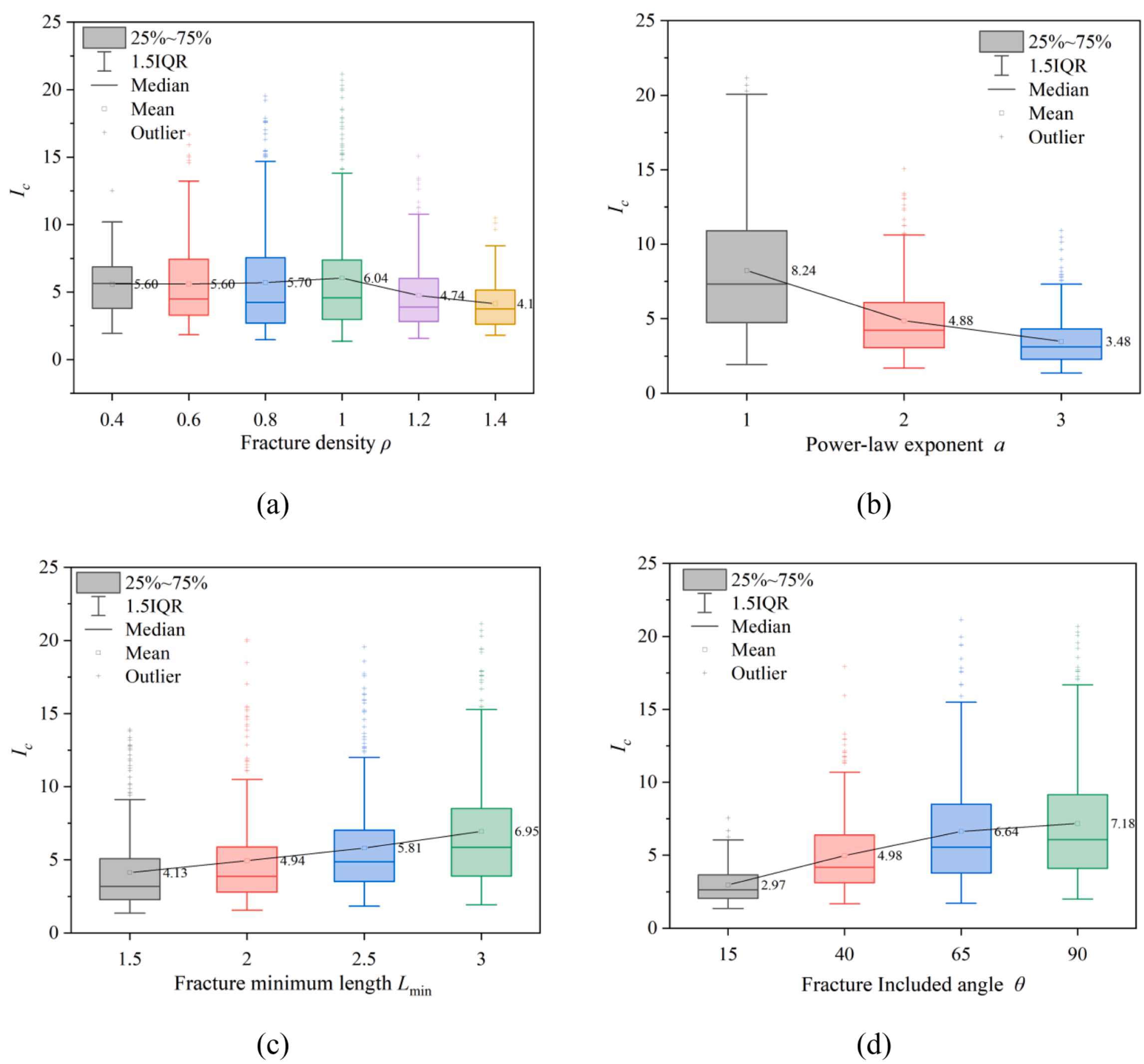


Fig. 6. The box plot of the effect of fracture density, power-law exponent, minimum fracture length, and included angle on connectivity index I_c.

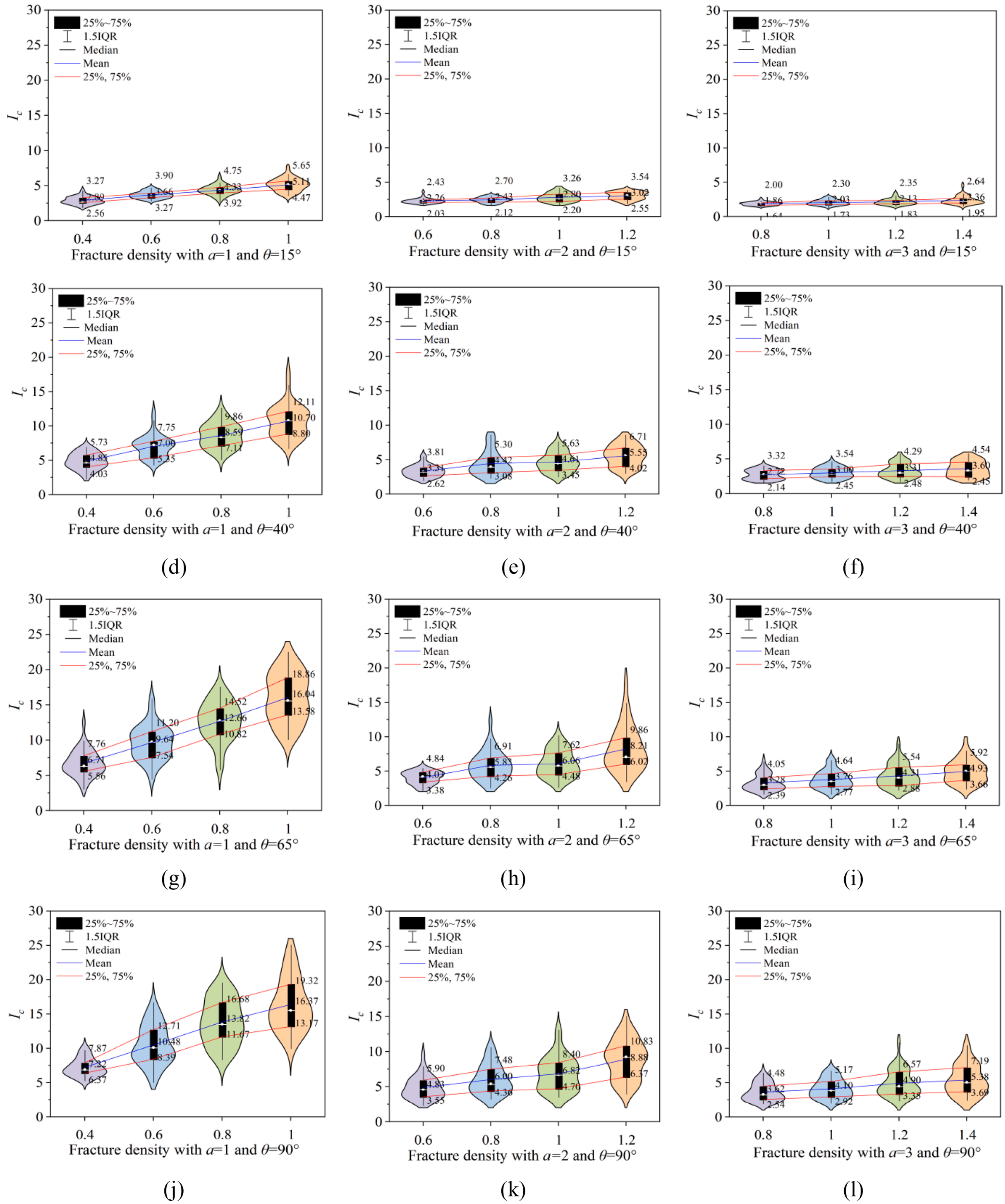


Fig. 7. The Violin Chart of the effect of fracture density on I_c under different a and θ.

network. It provides a tool for evaluating the hydraulic equivalence within the network, supplying valuable insights into the overall connectivity and flow characteristics of the fractures (Leung & Zimmerman, 2012). The definition of η is as follows:

$$\eta = \sqrt{A\varepsilon_{\mathrm{mean}}\rho_{\mathrm{seg}}}$$

$$\varepsilon_{\mathrm{mean}} = \rho_A\left(\frac{\bar{l}}{2}\right)^2 \tag{3}$$

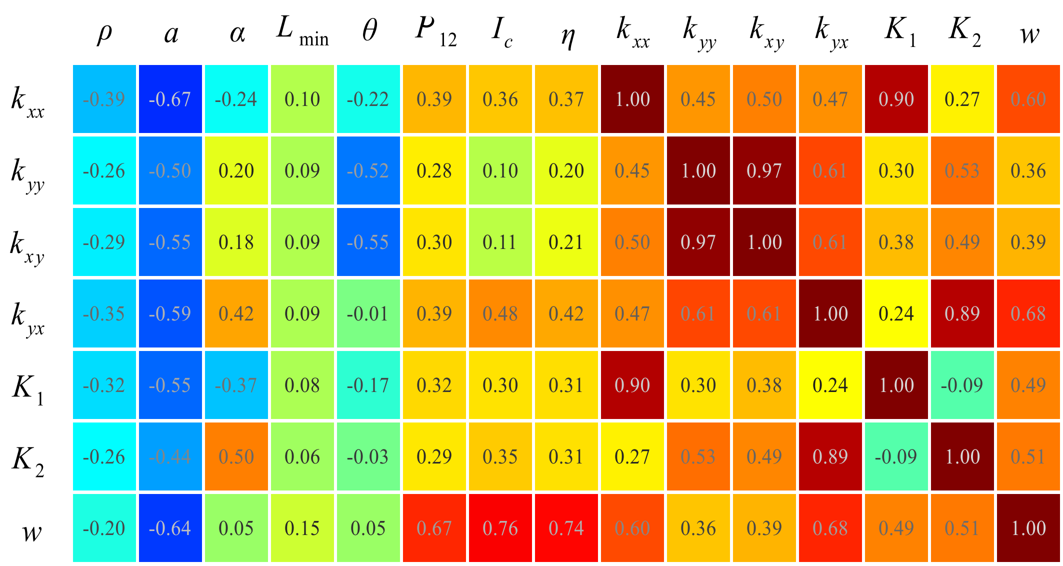


Fig. 8. The heat map of PPC between the fracture geometric features and the permeability.

$$\rho_{seg} = \frac{n_{seg}}{A} = \frac{N_F + 2N_{node}}{A}$$

where ε_{mean} and ρ_{seg} are the mean fracture density and segment density, and A is the domain area. The connectivity indices based on the same principle also include dimensionless density $\rho_A{}'$, Nc, and F_0 (Oda, 1986; Xiong et al., 2019). To a certain extent, these parameters are linearly correlated with fracture density P_{12}; thus, the specific analyses are ignored.

2.3. Finite element model of fracture seepage

The velocity field of water seepage can be calculated by modeling the incompressible steady seepage of the fracture network. Using a generalization of Darcy's law, the seepage velocity within the fracture can be expressed as:

$$u = -\frac{\gamma_w k_f}{\mu}\frac{\partial H}{\partial L} \tag{4}$$

where u is the velocity of fluid flow, μ is the dynamic viscosity of water, γ_w ($\gamma_w = \rho g$) is the unit weight of water, k_f is the fracture permeability, and $\partial H/\partial L$ is the hydraulic head drop. The correlation between hydraulic pressure drop $\partial P/\partial L$ and hydraulic head drop $\partial H/\partial L$ is:

$$\frac{\partial H}{\partial L} = \frac{\partial P}{\partial L \bullet \gamma_w} \tag{5}$$

Determined by cubic law, k_f can be expressed as:

$$k_f = \frac{d_f^2}{12 f_f} \tag{6}$$

where d_f is the hydraulic aperture, and f_f is the roughness coefficient.

The governing equation of single fracture can be expressed as (Jiang et al., 2013):

$$\frac{\partial u}{\partial L} = 0 \tag{7}$$

To solve the equation for the entire fracture network (Jafari and Babadagli, 2011, 2013), the flow must satisfy the following boundary conditions. For the water head boundary condition, we used linear pressure gradient boundary conditions, shown in Fig. 3:

$$P(0, y) = P_b \text{ left boundary}$$

$$P(l_x, y) = 0 \text{ right boundary}$$

$$P(x, 0) = P(x, l_y) = P_b(1 - x/l_x) \text{ upper and lower boundary} \tag{8}$$

For the flux boundary condition, the sum of inflow and outflow fracture flow should remain unchanged.

$$q_i = d_f u_i = 0 \tag{9}$$

Water primarily flows within the fracture network, leading to significant anisotropy in the permeability of fractured rock masses caused by the arrangement of fractures within rock formations. The permeability tensor serves as a crucial hydrogeological parameter that characterizes the permeable anisotropy of rock masses. Accurately calculating the permeability tensor is essential for analyzing fluid flow in fractured rock masses. In the two dimensions of the x-y coordinate system, the permeability K is represented by a symmetric, second-rank tensor (Durlofsky, 1991):

$$k = \begin{bmatrix} k_{xx} & k_{xy} \\ k_{yx} & k_{yy} \end{bmatrix} \tag{10}$$

The permeability can be obtained using low rate q_x and q_y values from Darcy's law (Chen et al., 2015). It can be written as:

$$q_x = -\left(\frac{k_{xx}}{\mu}\bullet\frac{\partial P}{\partial x} + \frac{k_{xy}}{\mu}\bullet\frac{\partial P}{\partial x}\right) = q_r \bullet n_x + q_u \bullet n_x + q_b \bullet n_x$$

$$q_y = -\left(\frac{k_{yx}}{\mu}\bullet\frac{\partial P}{\partial x} + \frac{k_{yy}}{\mu}\bullet\frac{\partial P}{\partial x}\right) = q_r \bullet n_y + q_u \bullet n_y + q_b \bullet n_y \tag{11}$$

where q_x and q_y are the flow rates along the x and y directions, respectively. q_r, q_u, and q_l are the Darcy velocities on the right, upper, and lower boundaries, and n_x and n_y are unit vectors along the x and y directions, shown in Fig. 3. The flow rate is equal to the sum of the product of each fracture aperture on the boundary and the linear velocity. q_r can be expressed as:

$$q_r = \sum_{i=1}^{N} v_{ri} \bullet b_{ri} \bullet n_x \tag{12}$$

2.4. Scenarios of two-dimensional synthetic fracture networks

Reeves et al. (2013) proposed that networks generated with mismatched fracture density and power law exponents may lead to unrealistic high-density networks. Therefore, to generate realistic fracture

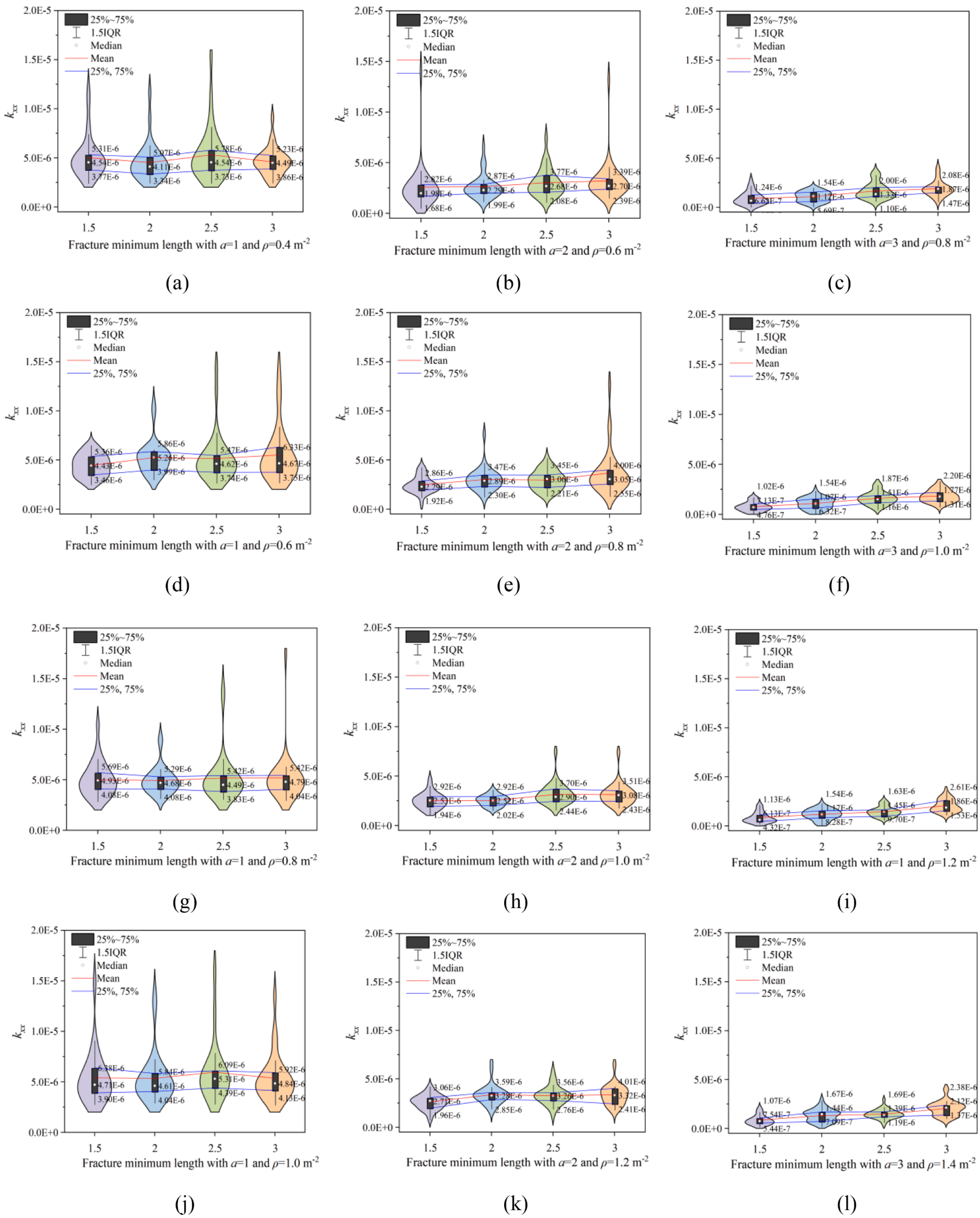


Fig. 9. The Violin Chart of the effect of fracture minumum length on k_{xx} under different a and ρ.

networks, a comprehensive experiment was pre-calculated to analyze connectivity. Finally, a series of 2D synthetic fracture networks were designed in a squared domain with 20 m × 20 m. Fracture locations were assumed to follow a Poisson distribution with variance $\lambda = 5$; fracture orientations α were modeled using a Fisher distribution centered around 0°, 22.5°, and 45° with a constant dispersion parameter $\kappa = 20$. The fracture set were assumed to be two groups; thus, the included angles θ between two group were set to 15°, 40°, 65°, and 90°. Fracture lengths were distributed based on a power-law distribution, power-law exponent a was assumed to be 1, 2, and 3. The fracture minimum Length $L_{\min}$ were set at 1.5, 2, 2.5, and 3 m, whereas upper bound $L_{\max}$ was 20 m. The fracture densities were designed based on corresponding exponent a, as shown in Table 2. The matrix permeability was assumed to be $k_m = 9{:}87 \times 10^{-16}$ m^2 (1 md), and the injection pressure is 20 Pa.

To mitigate the impact of fracture distribution randomness, each 2D model was generated three times using three random numbers. Finally, a total of 4 (four densities) × 3 (three power-law exponents a) × 4 (four

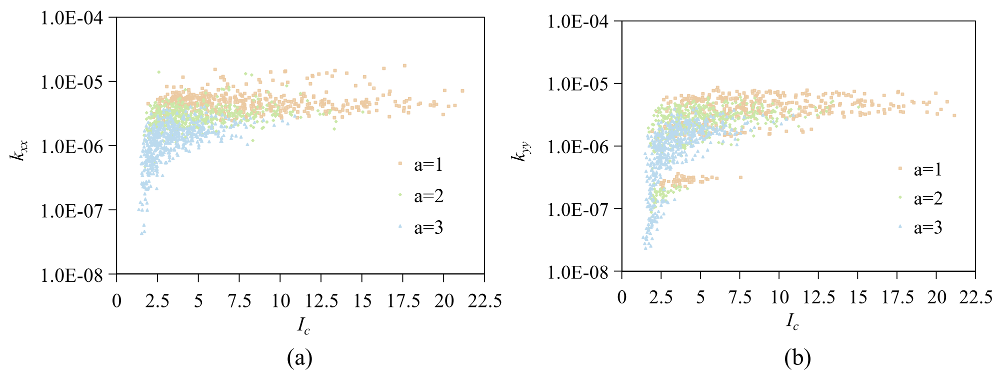


Fig. 10. The relationship between the permeability and connectivity index I_c: (a) k_{xx}, (b) k_{yy}.

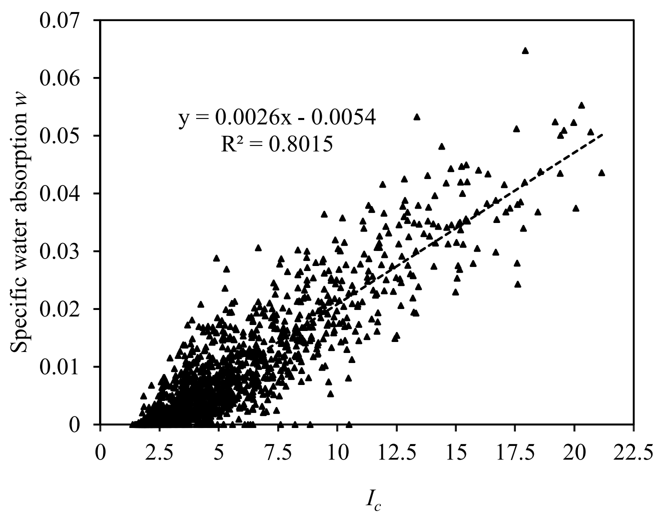


Fig. 11. Relationship between specific water absorption w and permeability, and I_c: (a) prediction results of K_1, (b) prediction results of K_2, (c) relationship between C_1 and I_c (d) relationship between C_2 and I_c.

orientations) × 3 (included angles) × 4 (four lengths) × 3 (three random numbers) = 1728 2D-DFN models were established. Each DNF model was simulated with three boundaries, along the x, y, and central radial directions, shown in Fig. 4. In total, 1728 * 3 = 5184 simulations were conducted.

3. Results and analysis

3.1. Fracture connectivity analysis of simplified DFNs

In the real word, the connectivity of fracture networks is usually overestimated. It is because that the isolated and dead-end fractures of initial DFNs increase the number and total length of fractures, but no contribution to the intersecting of fractures. A simplified DFN removing isolated and dead-end fractures can reflect the real connectivity and fluid flow paths in the process of fracture seepage. Thus, to deepen the comprehension of fracture network connectivity, a connectivity analysis of the simplified DFNs were conducted. Although the results in Section 2.1 have indicated that the connectivity after the simplification changes, it will not undermine the applicability of the study. This is because the geometric parameters of DNFs used to analyze the connectivity are initial and originate from the real fractured rock masses.

The Pearson Correlation Coefficient (PCC) method was employed to assess correlations between variables using Equation (13):

$$PCC_{m,n} = \frac{cov(m,n)}{\sigma_m \sigma_n} = \frac{E\left[(M-\mu_x)(N-\mu_y)\right]}{\sigma_m \sigma_n} \tag{13}$$

where σ_m, σ_n is the standard deviation of variables M and N, and cov(m, n) is the variable covariance. PCC$_{m,n}$ represent the correlation between variables M and N.

Fig. 5 displays the heatmap of the PCC between the fracture geometric features and the connectivity index. Analyzing the relationship between fracture geometric features, there is a notable correlation between fracture count density ρ and the power-law exponent a, while the correlation among other fracture geometric parameters appears relatively weak. This suggests a satisfactory independence among various indicators. The correlation between ρ and a is primarily attributed to the distinct ranges of ρ that were set for different values of a. In terms of fracture connectivity, a robust positive correlation exists among the connectivity indicators, yielding average correlation coefficients of 0.900, 0.905, and 0.951 for P_{12}, I_c and η, respectively. This indicates the effective characterization of complexity within the fracture network by each indicator. The relationship between connectivity and fracture geometric parameters reveals generally similar impact of different fracture geometric features on connectivity indicators. Considering the subsequent analysis of connectivity on the permeability of DFNs, I_c was ultimately selected as the connectivity indicator. It can be seen that the influence of fracture geometric features on I_c is ranked in the order of decreasing effect as follows: power-law exponent a (−0.55), included angles θ (0.47), minimum fracture length L_{min}(0.28), fracture count density ρ(−0.06) and fracture orientation α(0.02).

Specifically, Fig. 6 shows the fluctuations of I_c in various fracture features. It is evident that the a shows a negative correlation with I_c, resulting in an average I_c change from 8.24 to 3.48. As a increases, the long backbone within the DFNs gradually transforms into a shorter backbone, leading to the shortening of fractures length. Conversely, there is a positive correlation observed between the included angles θ and fracture minimum length L_{min}. An increase in θ results in a rise in I_c, causing an average I_c change from 2.97 to 7.18. More intersecting fractures contributing to additional fracture intersections and connectivity pathways. The impact of L_{min} is lower than that of θ. The increase in L_{min} resulted in an average I_c increasing from 4.13 to 6.95. The variation in ρ exhibits an initial upward trend followed by a plateau. This indicates that ρ affects I_c under the control of a and $L_{\min}$.

Additionally, further insights into the impact of key fracture geometric parameters on connectivity indicator I_c under different a, θ and ρ are analyzed, shown in Fig. 7. It can be observed that the influence of the a on I_c is greater than that of θ. Under the same θ, the growth rate of I_c decreases with the increase of a, even if a higher a corresponds to a higher ρ, and this variation remains substantial. For example, when $\theta = 15°$, the average I_c under the condition of ρ = 1.0 and a = 1 is 5.38, which is less than 7.22 of the I_c under the condition of ρ = 1.4 and a = 3. This indicates that, for real fracture networks,

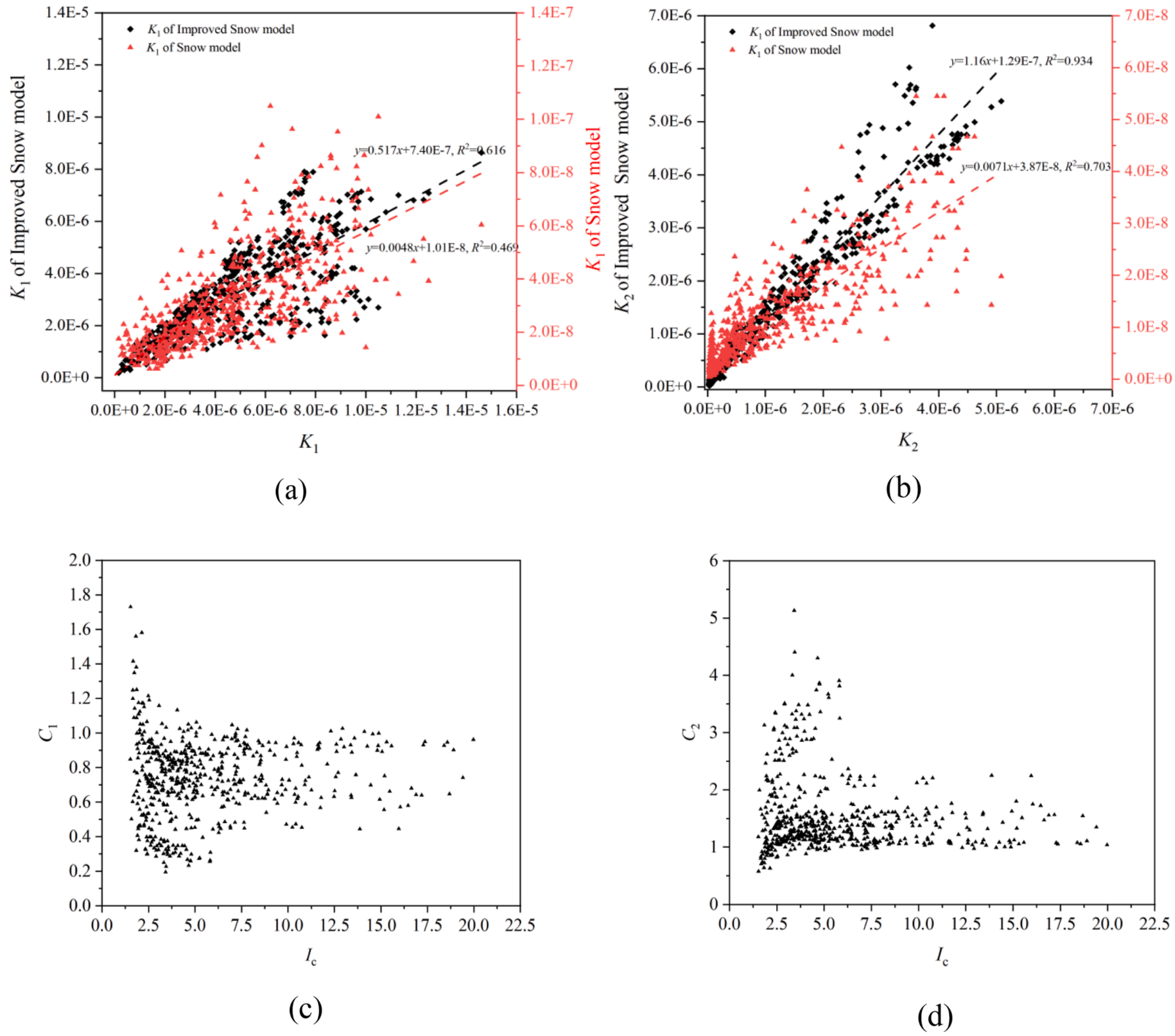


Fig. 12. Relationship between predicted and actual permeability, and I_c: (a) prediction results of K_1, (b) prediction results of K_2, (c) relationship between C_1 and I_c (d) relationship between C_2 and I.

under the condition of the same θ, the power-law exponent a play a more important role than Fracture count density ρ. Under the same a, the impact of θ on I_c is also significant, but this is mainly evident in the range of $\theta = 0^\circ \sim 65^\circ$. For instance, under the condition of $\rho = 1$ and $a = 1$, the average I_c is 5.11, 10.70, 16.04 for $\theta = 0^\circ$; 22.5°; 45°, respectively and the average I_c at $\theta = 90^\circ$ is 16.37, which only increases by 0.33, and the same situation occurs when $a = 2$ and $a = 3$. This implies that when the $\theta > 65^\circ$, the increase in θ cannot continuously increase the number of intersecting fractures, the crossings and flow paths between fractures have essentially stabilized. Therefore, as an evaluation parameter in the real world, more attention should be focused on situations where the a and the $\theta < 65^\circ$.

Furthermore, in an effort to predict the connectivity based on the actual fracture parameters, we consider the fracture count density ρ, power-law exponent a, fracture minimum length $L_{\min}$, and included angle θ and use multivariate linear function and exponential function to fit the connectivity I_c. The equation is as follows:

$$I_c = 1.846\exp(0.958D - 0.634a + 0.342L_{\min} + 0.011\theta), \quad R^2 = 0.8416 \tag{14}$$

3.2. Relationship between geometric features, connectivity, and permeability

3.2.1. Permeability

The permeability of DFNs is crucial for engineering geological applications, and it is influenced by various fracture geometry parameters. A comprehensive investigation of the influence of these fracture geometry parameters in permeability is the basis for realizing accurate predictions of permeability performance. Therefore, the correlation analysis among parameters related to fracture geometry features, connectivity, permeability (k_{xx}, k_{xy}, k_{yx}, k_{yy}), water absorption w and permeability principal values (K_1, K_2) is shown in Fig. 8.

Firstly, the heat map shows that the correlation between k_{xx}, k_{yy} and the power-law exponent a is most significant, followed by fracture orientation α and fracture density ρ. while the relationships with fracture minimum length $L_{\min}$ and included angle θ are weaker. It is observed that the impact of a on permeability k_{xx} and k_{yy} is analogous to its influence on connectivity, showing a decrease with an increase in a. However, it is noted that, in simulations, the hydraulic apertures are correlated with a. Hence, the effect of a on permeability is not solely attributed to the transition from a long fracture skeleton to a short one but also encompasses changes in hydraulic aperture size. When hydraulic aperture is unrelated to a, the control effect of A is weakened. Simultaneously, an unusual phenomenon can be observed, which is the

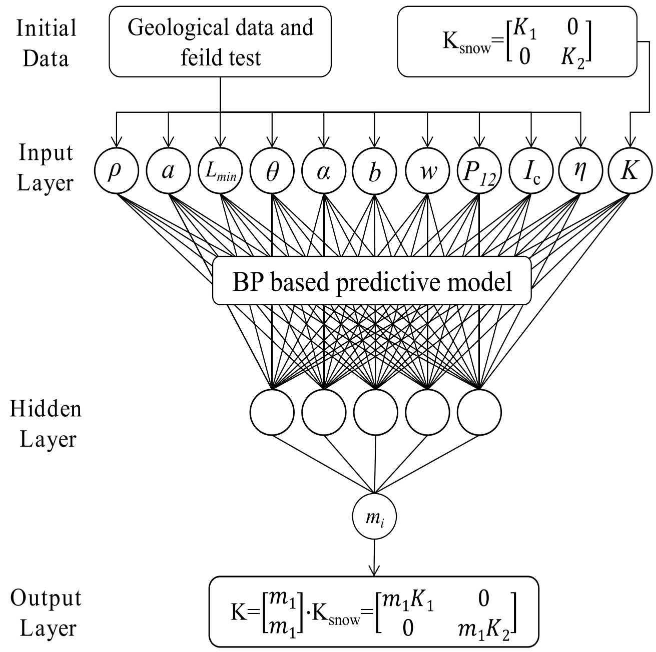


Fig. 13. Schematic diagram for a back-propagation model predictive model and its application.

negative correlation between permeability and fracture density ρ. The primary reason for this outcome lies in the setting of fracture network density ρ based on the power-law exponent a. The relationship between permeability (k_{xx} and k_{yy}) and connectivity shows a strong positive correlation. The average PPC between P_{12}, I_c, η, and k_{xx}, k_{yy} are 0.49, 0.55, and 0.51, respectively. It indicates that the connectivity index is significantly associated with the permeability (k_{xx} and k_{yy}) of the fracture network. Notably, the index I_c displays the most substantial correlation with permeability. Meanwhile, the water absorption w shows strong correlation with connectivity, which further demonstrate the application of connectivity in prediction permeability of DFNs.

Simultaneously, taking k_{xx} as an evaluation indicator for permeability, a detailed analysis of the variation in k_{xx} with $L_{\min}$ under different a and ρ conditions is shown in Fig. 9. It is observed that, with an increase in a, the k_{xx} of the DFNs gradually decreases. For instance, when $a = 1$, $L_{\min} = 3$ and $\rho = 0.4$, the average k_{xx} is 4.49e−6 m/s. However, when $a = 3$, $L_{\min} = 3$ and $\rho = 0.8$, the average k_{xx} decreases to 1.87E−6 m/s, marking a decline of 58.35 %. Importantly, this phenomenon does not diminish with an increase in ρ. Under conditions where a and $L_{\min}$ are the same, the k_{xx} shows a weak increase with an increase in ρ. Interestingly, it is observed that, compared to the $a = 1$, when $a = 2$ or 3, the growth trend of k_{xx} with the increase of ρ becomes more pronounced. For example, when $a = 1$, $L_{\min} = 3$, $\rho = 0.4$, the average k_{xx} is 4.49E−6 m/s, while the average k_{xx} is 4.84E−6 m/s when $a = 1$, $L_{\min} = 3$ and $\rho = 1.0$. It exhibits a growth rate of 7.80 %. Under similar conditions, the growth rates for $a = 2$ and $a = 3$ are 22.96 % and 13.37 %, respectively. Meanwhile, the variation of k_{xx} with $L_{\min}$ reveals that, as a increases, the impact of $L_{\min}$ on the growth rate of permeability in the fracture network gradually intensifies. For instance, when $a = 1$, no matter what the ρ is, the change of k_{xx} is not significant with $L_{\min.}$ However, when $a = 2$ or 3, the growth rates has become significant. Additionally, it is observed that with an increase in the a, the distribution range of permeability gradually narrows, and the number of anomalous permeability results diminishes.

In general, the combined influence of power-law exponent, fracture density, and fracture length indicates that the long skeleton in the fracture network predominantly controls the rock mass permeability. When the fracture orientation of the long skeleton deviates significantly from the flow direction, it alters the fluid flow path, leading to the extreme results in permeability along the flow direction. However, as short fractures occupy the fracture network, the control of permeability by fracture length gradually becomes apparent. The increase in the minimum fracture length enhances the likelihood of fracture intersections, leading to increased permeability. Furthermore, when fracture density increases, the growth in permeability is further enhanced. Therefore, in the assessment of fracture network permeability, the fracture length power-law exponent and the fractures orientation play an important role. In scenarios with a low power-law exponent, the roles of fracture density and minimum fracture length gradually become prominent. This analysis is based on 5148 simulation results, making it more convincing compared to studies with fewer factors and models.

Fig. 10 demonstrates the change of permeability(k_{xx} and k_{yy}) with connectivity. It shows a significant correlation exists between the I_c and the permeability. With an increase in connectivity, permeability k_{xx} and k_{yy} continuously increase, and gradually stabilized. Specifically, the permeability coefficient exhibits the fastest growth when $I_c < 6$. The amplitude of increase exceeds two orders of magnitude. Subsequently, as $I_c \geq 6$, the rate of increase gradually slows. When $I_c > 12$, the permeability coefficients for all directions do not exhibit significant influence. Therefore, I_c can be used as an indicator to evaluate permeability, but it is essential to consider the range of connectivity to reduce errors in the prediction process.

Meanwhile, due to the significant impact of the power-law exponent a on both permeability and connectivity, the relationship under different a is displayed. It can be seen that the power-law exponent a divide the relationship into three distinct regions. When $a = 1$, I_c ranges from 2.5 to 22.5, and the variation of k_{xx} with I_c is minimal, indicating a low correlation between connectivity and permeability, even when $I_c < 6$. When $a = 2$, I_c ranges from 1.5 to 15, and the correlation between k_{xx} and I_c strengthens. When $a = 3$, I_c ranges from x to x, and k_{xx} exhibits a nonlinear function with I_c. In Fig. 10b, it is evident that there are anomalous regions in the variation of k_{yy} with I_c. Considering the simulation conditions, these regions primarily occur under conditions where the fracture direction is perpendicular to the flow direction x.

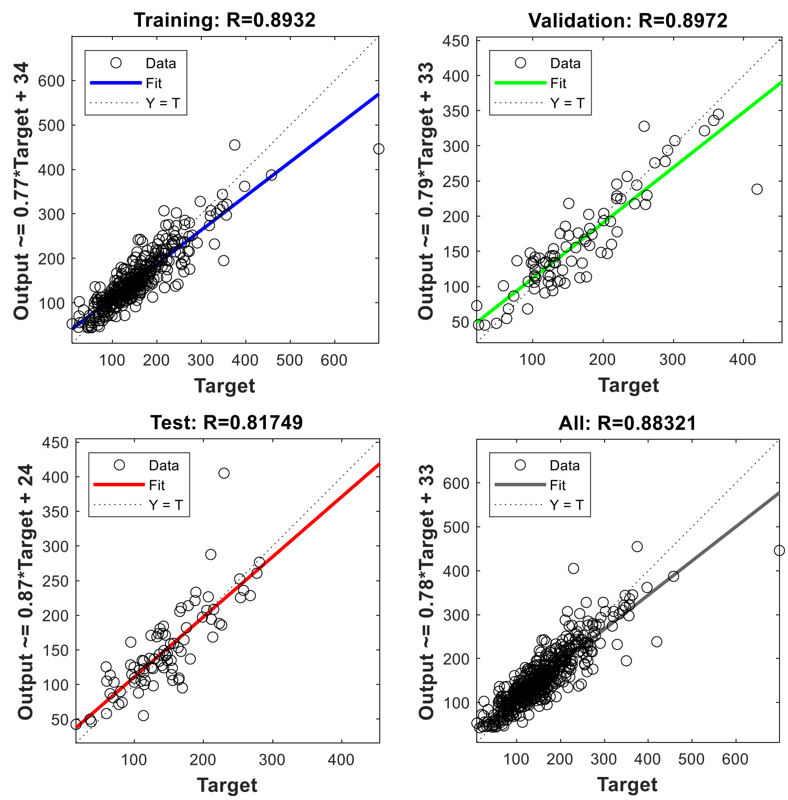
Fig. 11 shows the relationship between I_c and water absorption w. It can be seen that the fluid is injected from the center of the fracture network, and the relationship between w and Ic can be fitted using a linear function, the formula can be written as: $w = 0.0026I_c - 0.0054$, $R^2 = 0.802$. In practical engineering, we often evaluate the permeability of DFNs by water absorption w obtained through water injection tests. Combing the simulation results, it can be seen that if the permeability anisotropy of the fracture network is not our focus, the connectivity I_c can well reflect the overall permeability of the DFNs. However, for engineering projects that focus on the direction or the shape of water seepage, it is difficult to simply use connectivity to evaluate, such as grouting sealing in fractured rock mass or CO_2 storage.

4. Permeability predicted models of DFN

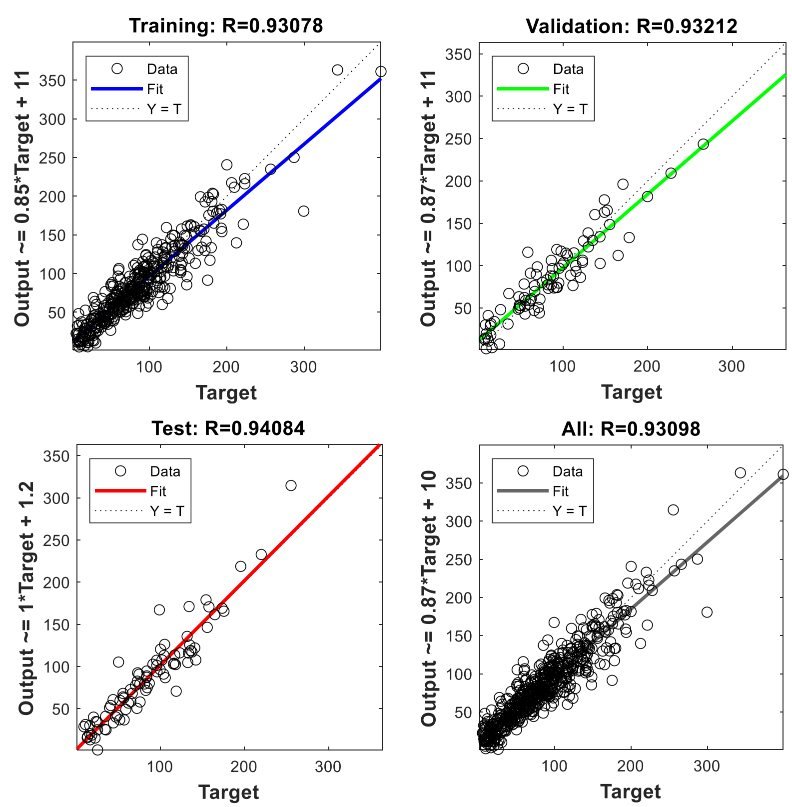
Building a model for predicting the permeability tensor of fracture rock mass based solely on readily available geological data, without knowing details of fracture distribution in rock mass, would be of great significance, as the permeability is important not only to the field of hydraulics but also to other areas. For example, in field grouting or geothermal development, it would be useful if the permeability tensor is known in advance, as it can be used to evaluate the diffusion range and extraction efficiency, which is critical for predicting the penetration range, plugging effect, and cost of underground engineering. Therefore, we attempted to establish a predicted model using theoretical models and artificial intelligence algorithms.

4.1. Snow model

Snow (1969) theoretically derived a formula for calculating the



(a)



(b)

Fig. 14. Correlation coefficient of predictive model developed based on BP model: Target (T) is the real m_i; output(Y) is the predicted m_i, (a) predicted of m_1, (b) predicted of m_2.

permeability tensor of fractured rock masses, assuming the orientation of the i-th group of fractures as α_i. The unit vector of the i-th group of fractures can be defined as follows:

$$ni = (nxi, nyi) \tag{15}$$

Based on spatial analytical geometry, the relationship between the vector components and the fracture occurrence is given by:

$$\begin{cases} n_{xi} = \sin\alpha_i \sin\beta_i \\ n_{yi} = \cos\alpha_i \sin\beta_i \end{cases} \tag{16}$$

The permeability tensor of i-th fracture group can be calculated according to Eq. (17):

$$K_i = \frac{gb_i^3}{12\mu s_i}\begin{bmatrix} 1 - n_{xi}n_{xi} & -n_{xi}n_{yi} \\ -n_{yi}n_{xi} & 1 - n_{yi}n_{yi} \end{bmatrix} = \begin{bmatrix} k_{i-xx} & k_{i-xy} \\ k_{i-yx} & k_{i-yy} \end{bmatrix} \tag{17}$$

where b_i is the average fracture aperture of i-th group, μ is the dynamic viscosity of water, g is gravitational acceleration.

If there are N sets of fractures in the rock mass, the permeability tensor K is identical to the sum of each permeability tensor:

$$K = \sum_{i=1}^{N} K_i \tag{18}$$

Meanwhile, the permeation tensor principal values (K_1, K_2) and their principal directions can be obtained from the permeation tensor by solving the eigenvector and eigenvalue of the second order symmetric matrix.

$$K = \begin{bmatrix} K_1 & 0 \\ 0 & K_2 \end{bmatrix} \tag{19}$$

Fracture average aperture plays an important role in predicting the permeability, usually measured through field investigations. However, when the average fracture aperture has a significant correlation with fracture length, the average aperture will be difficult to measure. In this paper, a power-law function is used to describe the fracture aperture b of each fracture. Thus, the average fracture aperture can be obtained using the following equation:

$$\overline{b} = \sum_{i=1}^{n} \frac{b_i}{n} \tag{20}$$

where n is the total number of fractures, b_i is the fracture aperture of the i-th fracture.

As previously noted, the fracture aperture is notably affected by the fracture length, which means that it is most closely related to the parameters L_{min} and a. Thus, a relationship between the average fracture aperture, L_{min} and a is established by fitting the simulation results. Due to the performance of accuracy in the fitting process, a paraboloid function, $z = z_0 + ax + by + cx^2 + dy^2$, was selected. The fitting equation is as follows:

$$b = 0.0004 - 0.0002a + 5.28 \times 10^{-5}L_{min} + 2.25 \times 10^{-5}a^2 - 2.72 \times 10^{-6}L_{min}^2 \tag{21}$$

Combining the Eq. (16)–Eq. (21), the permeability tensor calculated by initial Snow model can be obtained.

However, many studies has proven that the initial Snow model has an obvious error in predicting permeability of discontinuity fracture network. Therefore, the corrected parameter m is introduced, which can be expressed:

$$m = K_w / K_0 \tag{22}$$

where K_W is the equivalent permeability coefficient can be obtained according to the the specific water absorption w. It can be calculated based on Babushkin's formula:

$$K_w = 0.525wlg\frac{cL}{r} \tag{23}$$

where w is the specific water absorption, L/min·m·m. It is identical to $0.01q$, where q is the permeable rate, Lu; L is injection length, set at 1 m, c is the coefficient, set at 1.32, and r is the radius of the hole.

K_0 is the geometric average of the two principal values of the permeability tensor, which can be written as:

$$K_0 = \sqrt{K_1K_2} \tag{24}$$

The corrected permeability coefficient tensor for fractured rock masses can be obtained by taking the correction coefficient m into the permeability coefficient tensor:

$$\overline{K} = m\begin{bmatrix} K_1 & 0 \\ 0 & K_2 \end{bmatrix} = \begin{bmatrix} mK_1 & 0 \\ 0 & mK_2 \end{bmatrix} \tag{25}$$

Fig. 12 shows the relationship between the predicted principal values of the permeability tensor obtained by initial Snow-model, improved Snow-model and actual principal values. It can be seen from Fig. 12a, although the K_1 calculated by initial Snow-model and improved Snow-model both shows positive correlation with actual K_1, there is an error of two orders of magnitude of initial Snow-model compare with the actual K_1. It demonstrates the limitation of initial Snow-model in prediction principal values of the permeability tensor in discrete fracture network. The same situation also occurred in the prediction results of K_2 in Fig. 12b. Conversely, the results calculated by improved Snow-model has a good description. The R^2 of K_1 and K_2 is 0.616 and 0.934, respectively.

Fig. 12c and d shows the ratio C_i of the results calculated by improved Snow-model to the actual values. It can be seen from Fig. 12c, most of C_1 less than 1.0, the range is mainly concentrated between 0.6 and 1.0. It demonstrates that the predicted results K_1 are less than the actual value. With the increase of I_c, the DFNs becomes more connection, the C_1 tends to stabilize and the K1 tends to approach the actual value. Fig. 12d shows the ratio C_2 changes with I_c. Conversely, most of C_2 more than 1.0. The differences between C_1 and C_2 illustrates that although we have corrected the permeability principal value, this correction is isotropic and still cannot consider the impact of direction, which may result in values expanding or decreasing in different directions.

4.2. Back-propagation model

The characteristic parameters of rock fractures can be obtained from geological data, and the potential injection flow rate can be read and determined from the pump, which are all data that can be used. Through data analysis, we found that a predictive model of high accuracy can be built utilizing only geological data and the correlated injection rate based on a neural network algorithm developed by the Matlab Toolbox App. The detailed algorithm and principle have been introduced in previous papers. In this study, 11 nodes were used in the input layer, with 8 hidden nodes in the hidden layer and 2 output nodes in the output layer. The ten nodes are ρ, a, L_{min}, θ, α, b, w, P_{12}, I_c, η and k_i, respectively. k_i is the permeability principal value calculated by the initial Snow-model. Two output layer nodes are correction factors m_1 and m_2, respectively. After the algorithm, the predicted k_i value can be obtained by Eq. (24), shown in Fig. 13. In Fig. 13, it is observable that the Snow model, which considers the effects of fracture orientation and geometric features, is integrated into the BP model, with each geometric feature of the fractures also included in the model. There are interrelationships among these physical parameters. However, unlike traditional data analysis methods, for data mining models, collecting extensive data is acceptable, even if there may be interrelationships between these parameters. This is because that by gathering these parameters, more details can be retained for a deeper understanding of the permeability of fractures and ultimately improve predictive capability.

The predicted model was developed using simulation data. The average results of each parameter can used. In total, there are 576 fracture models, and the total sample size is 576. All samples were used for developing the predictive model, and they were divided randomly into three parts: training set (70 %, 403 samples), validation set (15 %, 86 samples) and test set (15 %, 86 samples), where the training and validation sets work together to build a predictive model.

The performance of the model is shown in Fig. 14. The correlation coefficient R is used to evaluate the error. As shown in Fig. 14a, the R of m_1 is 0.8932, 0.8972, 0.8175 and 0.8832 for training, validation, test and all. The R of m_2 is 0.9308 0.9321, 0.9408 and 0.9310 for training, validation, test and all, shown in Fig. 14b. It indicates that the target can be accurately described by the output. The model is described using the neural network algorithm based predictive model utilizing only geological data and the corresponding inlet rate.

5. Conclusion

This study proposed a method to realize a large number of seepage numerical simulations from modeling and parameter calculation. A total of 1728 fracture models were established, and based on the models, 1728 * 3 = 5428 simulations were conducted for x axis, y axis, and center radial diffusion boundary conditions. The theoretical analysis, empirical formulas, and data statistical analysis of the simulation results were used to discuss the relationship between fracture network characteristics, connectivity, water absorption, and permeability. The following are the study's key findings:

(1) The influence of fracture geometric features in connectivity I_c are investigated. Power-law exponent and included angle have notable correlations with I_c. In the real DFNs, more attention should be focused on situations where the Power-law exponent of fracture length and the included angle less than 65°
(2) The analysis of permeability demonstrates that the permeability of DFNs is controlled by power-law exponent of fracture. The influence of fracture density and fracture minimum length varies with the increase of a. When $a = 1$, the influence of fracture density and minimum crack length on permeability is relatively weak. When $a \geq 2$, the influence of fracture density and fracture minimum length gradually intensifies.
(3) The connectivity I_c has a strong correlation with water absorption. In practical engineering, if the permeability anisotropy of DFNs is not our focus, the connectivity I_c and water absorption can well reflect the overall permeability of the DFNs.
(4) The theoretical and neural network models of permeability in a fracture network are established separately. The results show that owing to the network discontinuity, the initial Snow-model models have significant error. The improved Snow-model can describe the permeability, but it still cannot consider the impact of flow direction. The neural network algorithm model is a good choice, which can accurately describe the permeability of networks only considering the geological data and specific water absorption.

The major weakness of this study, however, is that we only focus on the two-dimensional random fracture network and considers the scenario involving two sets of fractures. These limitations may potentially lead to discrepancies between the simulated and actual behaviors. In future research endeavors, we plan to expand the application of proposed method to tackle more complex three-dimensional rough fracture simulations. Additionally, we will conduct focused case analyses to investigate the permeability properties of rock masses in specific regional area.

CRediT authorship contribution statement

Chenghao Han: Writing – original draft, Methodology, Funding acquisition. **Shaojie Chen:** Writing – review & editing, Resources, Project administration. **Feng Wang:** Writing – review & editing, Supervision, Conceptualization. **Weiye Li:** Visualization, Software, Formal analysis. **Dawei Yin:** Writing – review & editing, Methodology, Conceptualization. **Jicheng Zhang:** Visualization, Software, Investigation. **Weijie Zhang:** Software, Data curation, Conceptualization. **Yuanlin Bai:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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